The Assignment Submission Portal

Week 8 Coding Assignment - Question 2

Design: Student 1--0..* Submission; Assignment 1--0..* Submission. Assignment is abstract; CodingAssignment and WrittenAssignment define different daily penalty rates.
Java solution:

```
import java.time.*; import java.time.temporal.ChronoUnit;
abstract class Assignment {
final String title; final int max; final LocalDate due;
Assignment(String t,int m,LocalDate d){title=t;max=m;due=d;}
abstract double dailyPenalty();
double finalMarks(double awarded,LocalDate submitted){
long late=Math.max(0,ChronoUnit.DAYS.between(due,submitted));
return awarded*Math.max(0,1-late*dailyPenalty());
}
}
class CodingAssignment extends Assignment { CodingAssignment(String t,int m,LocalDate
d){super(t,m,d);} double dailyPenalty(){return .10;} }
class WrittenAssignment extends Assignment { WrittenAssignment(String t,int m,LocalDate
d){super(t,m,d);} double dailyPenalty(){return .20;} }
class Student { final String name; Student(String n){name=n;} }
class Submission {
enum Status { Submitted, Graded }
final Student student; final Assignment task; final LocalDate submitted;
private Status status=Status.Submitted; private Double marks;
Submission(Student s,Assignment a,LocalDate d){student=s;task=a;submitted=d;}
Status status(){return status;}
double grade(double awarded){if(status!=Status.Submitted)throw new
IllegalStateException("Already graded");
if(awarded<0||awarded>task.max)throw new IllegalArgumentException("Invalid marks");
marks=task.finalMarks(awarded,submitted);status=Status.Graded;return marks;}
boolean canResubmit(){return status!=Status.Graded;}
}
public class Main { public static void main(String[] args){
Student a=new Student("Asha"),r=new Student("Ravi");
Assignment c=new CodingAssignment("Linked List Lab",50,LocalDate.of(2026,3,10));
Assignment w=new WrittenAssignment("Design Essay",50,LocalDate.of(2026,3,12));
Submission s1=new Submission(a,c,LocalDate.of(2026,3,10));
Submission s2=new Submission(r,w,LocalDate.of(2026,3,14));
System.out.println("Asha’s submission for Linked List Lab received (on time). Status:
"+s1.status());
System.out.println("Ravi’s submission for Design Essay received (2 days late). Status:
"+s2.status());
System.out.println("Asha graded: "+s1.grade(45)+"/50. Status: "+s1.status());
System.out.println("Ravi graded: "+s2.grade(40)+"/50 after 40% late penalty. Status:
"+s2.status());
if(!s1.canResubmit())System.out.println("Cannot resubmit: Linked List Lab has already been
graded.");
} }
```

Expected key results: Asha 45/50; Ravi 24/50 after two late days; status changes to Graded. Submission owns its status; callers cannot modify it directly. Penalty rules belong to Assignment subtypes.